Waddington: A Constrained Conversational Agent with Cross-Experiment Priors for Sequential CRISPR Gene Selection*

Method, results, and figures

Abstract

Designing a genome-scale CRISPR screen means choosing, over a few rounds of ∼128 perturbations, which genes to test — a batch active-learning problem. We present Waddington, a hybrid that fuses an online-retrained gradient-boosted ranker with a tool-less LLM, reaching hit@R5 = 0.253 across nine public screens under a configuration fixed *without target-screen outcomes* — a global fusion weight selected on non-target screens, a metadata-only target feature policy, and no privileged anchor features — above online ML alone (0.224), a padded LLM baseline (0.222), and a diversity baseline (0.134). (An earlier target-aware routed variant scores 0.259, with no detected difference; we report the leakage-free system.) To place this against prior work we ran both relevant lineages on our own benchmark under identical evaluation: BioDiscoveryAgent's LLM agent scores 0.105 and GeneDisco's active-learning acquisition functions 0.113 on the six shared screens — yet our leave-one-out prior, a static model with *no* LLM, reaches 0.187, beating both. The differentiator is therefore a cross-experiment prior, not the agent: both prior methods learn only within a single screen. That prior is largely a *recurrence* prior — a simple cross-screen hit-frequency ranking matches the learned model (its signal is recurrent hits and generic essentiality, not novel phenotype-specific biology). **Against its matched static prior, Waddington improves hit@R5 from 0.231 to 0.253 (+0.022, 95% CI $[0.012, 0.036]$); but replacing that prior with the pure hit-frequency ranking yields no advantage over recurrence alone (0.239 vs. 0.243).** Counterfactual attribution places the LLM as a *verifier*, not a proposer: after adjusting for the ML score, endorsement still predicts hits (adjusted odds ratio 1.68, 95% CI $[1.05, 2.52]$), concentrated among top-tertile ML candidates (43.1% vs. 17.4%), while giving it tools, explicit memory, or free choice does not help in our setup. Finally, anonymizing gene names and supplying structural features instead gives similar mean performance (no detected difference, also without the privileged anchor features), consistent with structure-sufficiency rather than LLM reasoning per se — though Steinhart still benefits from gene-name semantics and we cannot separate biological knowledge from memorisation. We ship the anonymized mode as an opt-in.

All numbers in this document are computed directly from the frozen benchmark outputs (the JSON files under `workspace/results/sequential/`) by `waddington_select.analysis`; no value is transcribed by hand. Figures are produced by `python -m waddington_select.analysis figures`.

*We keep the term *agent* deliberately: Waddington runs a stateful sequential loop, observes experimental feedback each round, updates its actions across rounds, and is driven through a natural-language conversational interface with the researcher. *Constrained conversational* signals what it is *not* — a freely planning tool-using agent. Its policy deliberately confines the LLM to *verifying and re-ranking* candidates proposed by a calibrated ML model rather than choosing genes on its own. Supporting experiments further show that, under the tested action schema and budget, a freely planning tool-using variant underperforms the constrained policy (Appendix).

Related work

Two lines of work bear directly on ours. **Active-learning benchmarks for experimental design.** GeneDisco (Mehrjou et al., 2021) frames screen design as batch active learning and evaluates acquisition functions — uncertainty, margin, coreset, k-means, BADGE — over a surrogate model retrained on the genes tested so far. **LLM agents for biology.** BioDiscoveryAgent (Roohani et al., 2025) and PerTurboAgent prompt a large language model to plan rounds and name genes, optionally with retrieval / pathway-enrichment tools. What both share is that they learn *within a single experiment*: the surrogate, or the LLM's in-context feedback, starts from scratch on each screen. Our contribution is orthogonal to the choice of acquisition rule or LLM: a *cross-experiment* supervised prior (leave-one-out over related screens) combined with the LLM used as a *verifier* of that prior's candidates rather than as a free agent. Running both lineages on our benchmark under identical evaluation (Table 2) places them together at $\sim$0.11, against our prior's 0.187 — so the gain we report over their published numbers should be read as "a transferred prior helps a lot", not "a better agent". We deliberately give the LLM no tools, a choice the attribution analysis (Explainability, below) justifies rather than assumes.

Method

Problem: sequential batch selection

A screen exposes a pool of $N \approx 18{,}000$ genes whose hit labels are hidden. In each of $R = 5$ rounds the agent selects a batch of B genes ($B = 128$; $B = 32$ on the two small screens, Scharenberg22 and K562-Essential), an oracle reveals hit / non-hit for exactly those genes, and the agent adapts before the next round. The objective is the round-5 *hit ratio*

$$\text{hit@R5} \;=\; \frac{|\{\text{unique hits among the } 5B \text{ selected genes}\}|}{\text{total hits in the screen}},$$

i.e. how much of the screen's biology is recovered on a fixed budget. The loop is simply *select* $\rightarrow$ *reveal* $\rightarrow$ *update*, repeated five times; everything below concerns how a round's batch is chosen.

Gene features and the cross-experiment prior

Each gene carries two families of features. **Gene-intrinsic** features are constant across screens: STRING network degree and hubness, total PPI weight, the pLI loss-of-function constraint, and KEGG / Reactome pathway counts. **Anchor-relative** features measure a gene's proximity to the *phenotype's anchor genes* (a small seed set describing the screen's biology): PPI connectivity to the anchors (`g1_ppi_score`), ARCHS4 co-expression with them (`archs4_coexpr`), and KEGG pathway overlap (`kegg_overlap`). To these nine we append knockout-derived **DepMap** essentiality features. Anchor-relative features are modality-agnostic — they depend on the phenotype's biology, not on the direction of perturbation — a property the reason-vs-recall analysis (Table 7) relies on.

The reported system uses only the gene-intrinsic and metadata-selected DepMap features; it excludes the anchor-relative features entirely. A leakage audit later found that 51% of the anchors are themselves target hits (privileged information, §), so the anchor-relative family is dropped from the final configuration and retained only in the legacy predecessor and the reason-vs-recall probe. Read the anchor features below, therefore, as historical candidates, not part of the reported prior.

The **cross-experiment prior** is a leave-one-out (LOO) LightGBM classifier: to score a target screen it is trained on the *other* screens' hit labels using the screen-appropriate subset defined by the final feature policy — gene-intrinsic features and metadata-selected DepMap features, with no anchor-relative features — with `scale_pos_weight` correcting the ∼5% class imbalance (300 trees, learning rate 0.05, 31 leaves). This prior — with no LLM and no in-experiment data — is what the external comparison (Table 2) shows already exceeds the strongest single-experiment baseline by ∼0.08, because it transfers labels across screens that GeneDisco and BioDiscoveryAgent cannot.

The C-arm: online ML and an LLM, fused

Each round the C-arm forms a batch by combining two components that see the same revealed feedback.

Component A — online ML. An adaptive LightGBM ranker. In round 1 it *is* the LOO prior (no in-experiment data yet); from round 2 it is retrained on the LOO data *plus* the in-experiment revealed (gene, label) pairs, once at least 8 genes have been revealed, and re-ranks the whole pool by predicted hit probability. In the legacy predecessor's paired ablation, removing this online retraining produced the largest average loss (Appendix, Table 11).

Component B — LLM endorsement (candidate verification). A tool-less LLM (Claude Haiku 4.5, temperature 0) is prompted with the task description, the top-4 most relevant distilled lessons from a cross-experiment memory of past screens (ranked by `_rank_memory_by_relevance`, and never including the target screen), and the revealed hit / non-hit history so far. It names genes; these are matched against the pool, and any shortfall is padded from the static LOO ranking — padding that is tracked and never credited to the LLM (Table 4[†]).

Fusion. The two components are combined by a weighted score

$$\mathrm{score}(g) \;=\; w_{\mathrm{ML}} \cdot \mathrm{ml}(g) \;+\; w_{\mathrm{LLM}} \cdot \mathbf{1}[g \text{ named by the LLM}],$$

and the top B genes are taken. In the reported system the weight is a *single global constant*, not a per-screen choice.

The final configuration: fixed without target-screen outcomes

The system we report fixes everything a screen needs *before* it runs, and never uses the target's realized labels. Two ingredients: a global fusion weight, selected once on the *non-target* screens (below), and a target feature policy that is a pure function of pre-experiment metadata — pool size, perturbation modality, cell line.
Fusion weight. A single global $w_{\mathrm{LLM}} = 0.2$ ($w_{\mathrm{ML}} = 0.8$) on every screen. That value is chosen once, by leave-one-out over the *other* screens (it wins in every fold); no per-screen routing and no hit rate are involved.
Feature policy, a pure function of experimental-design metadata:

Metadata of the screen	DepMap features
gain-of-function (CRISPRa) or curated-essentiality panel	**none**
cell line = K562	pan-cancer only (drop K562-matched)
otherwise	pan-cancer + K562-specific

DepMap is knockout-derived, so it misleads on a gain-of-function screen; a pan-cancer essentiality prior nearly restates a curated-essentiality label; and a cell-line-matched essentiality feature is a proxy label on its own cell line. SHAP confirms the model uses these features where kept and ignores them where dropped (Table 12). **The anchor-relative features are dropped entirely**: a leakage audit found 51% of anchors are themselves target hits (privileged information), and removing them costs only 0.009 (§).

A legacy target-aware router (analyzed in §, not used here). An earlier version set w_{LLM} per screen from the *realized* hit rate (`ml_heavy` 0.8:0.2 / `two_stage` / `baseline` 0.6:0.4) and kept the anchor features. It scores 0.259 (no detected difference from the final system) but reads a test-set statistic, so we retain it only as a robustness comparison. The detailed ablations, attribution, and SHAP were computed on this near-performing predecessor and are kept as supporting analyses; we did not re-estimate the individual component effects under the final configuration, and do not assume they all transfer quantitatively.

Why the reported system avoids the legacy router. The legacy route reads the realized hit rate — a statistic of the test set, determining the route on seven of the nine screens — and keeps the privileged anchor features. That is precisely why the reported system uses neither. Fixing the weight without target-screen outcomes (on the training screens) and the feature policy from pre-experiment metadata, and dropping the anchor features, reaches 0.253, with no detected difference from the routed 0.259 (paired -0.006, CI $[-0.019, 0.007]$). It improves over its *matched* static prior by +0.022; relative to the differently-configured benchmark LOO baseline the difference is +0.036 (§). The main external claim never used the router anyway: the cross-experiment prior that beats single-experiment active learning and LLM agents (0.187 vs. ≈ 0.11, Table 2) is the routing-free static LOO-LightGBM. We keep the legacy variant only because the detailed ablations, attribution, and SHAP were computed on this near-performing predecessor and are retained as supporting analyses.

Design principle. Throughout, the LLM *re-weights a calibrated model's candidates* rather than choosing genes freely. This is deliberate: the attribution analysis (Explainability, below) shows the LLM is a strong *verifier* of the ML's picks but a weak *proposer* on its own, which is why giving it more agency (its own tools, or letting it choose unaided) does not help.

The algorithm, and one worked round

```
Input:  pool G, rounds R, batch B, screen s
Global fusion weight (w_ml, w_llm):  preselected on NON-target screens (leave-one-out)
Target feature policy:  from pre-experiment metadata (modality, cell line); no anchor feats
Prior p0 = LOO-LightGBM fit on OTHER screens' (features, label)    // no target labels
selected = {}
for t = 1..R:
  ml = online_ranker.scores(G)    // t=1:  p0;  t>1:  retrained on revealed pairs (≥8)
  named = LLM.pick(task, memory[:4], revealed_history)  // gene NAMES only; no ml score
  score(g) = w_ml·ml(g) + w_llm·[g ∈ named]
  batch = top_B(score, exclude=selected);  pad from static LOO if short  // pad not credited
  revealed = oracle.reveal(batch);  selected ∪= batch;  online_ranker.add(revealed)
Output:  hit@R5 = |unique hits in selected| / total_hits(s)
```

A worked round (IFN-γ, round 1). Route `ml_heavy` (w_{ML}=0.8, w_{LLM}=0.2), B=128; the ML ranks all 17,785 genes and the LLM independently names 118. The fused top-128 is 16 *both* (ML top-k *and* LLM-named), 97 *ML-only*, 15 *LLM-only*. Outcomes: the *both* bucket hits 14/16 (88%; e.g. ZAP70, VAV1, GRAP2, all ML≈0.98), ML-only 37/97 (38%; CD5, CD247, MAP4K1), LLM-only 6/15 (40%; the LLM lifted mid-ranked genes such as TSC2 and RHOA, ML≈0.82, into the batch). The round finds 57/128 hits. In one round you can see the whole thesis: the batch is ML-dominated, and the LLM's mark concentrates hits in the small bucket both endorse. (This single round is a mechanistic illustration, not a representative rate: its 88% agreement hit rate runs above the 46.5% pooled agreement rate of Table 8; the pattern, not the level, is the point.)

The conversational interface

A scientist drives Waddington in natural language rather than by editing config. A lightweight intent router (itself an LLM call) parses each free-text message into a structured action — `suggest` one batch, run an interactive `experiment` campaign, `simulate` end-to-end, `register` a new phenotype, or `chat` — and extracts the phenotype, batch size, number of rounds, the run mode, and any genes the scientist reports as hits or non-hits in that message. An `experiment` then runs the round loop conversationally: the system proposes a batch, the scientist commits, outcomes are revealed — either by the benchmark oracle or, in *upload* mode, by the scientist supplying their own screen readout (a MAGeCK or CSV file, from which hits are derived) — and the revealed hits and non-hits become cross-round state that conditions the next proposal. Biology outside the nine benchmark screens is handled by `register` (phenotype name, task, measurement, seed genes), after which the same pipeline ranks it.

Crucially, the conversational LLM is confined to *intake and narration*: every gene choice is made by the constrained selection policy above, never by the conversational chat model. So the interface adds natural-language control (goal, budget, round-by-round feedback, new phenotypes) without relaxing the ML-calibration constraint that the rest of this paper argues for. We describe this interface as a system component and do not benchmark it; features such as candidate veto or rationale-on-demand are natural extensions we have not yet built.

Table 1: The nine benchmark screens. The phenotype scores and hit sets are taken *unchanged* from the BioDiscoveryAgent (BDA) release (Roohani et al., ICLR 2025); we verified for all nine that our hit labels are exactly `topmovers` ∩ (pool \ CEGv2). Task descriptions are BDA's own task prompts, not our paraphrases. **We remove the 684 core-essential genes of CEGv2** (Hart et al.) from the candidate pool — BDA's own non-essential ("N/E") convention, since essential genes are detected as hits under almost any screen and reward nothing. Columns show the pool *before* → *after* that filter; all results use the filtered pool (the two K562 screens contain no CEGv2 gene, so they are unchanged).

Perturbation. KO = CRISPR knockout (loss of function); CRISPRi = interference/knockdown (loss of function); CRISPRa = activation (gain of function). Note that our DepMap features are *all knockout-derived*, so they transfer least well to the gain-of-function screen (Steinhart) — which is exactly where the LLM contributes most (Table 11). *The Schmidt IL-2 and IFN-γ data are the *CRISPRi* (loss-of-function) arm. Schmidt et al. ran both CRISPRa and CRISPRi genome-wide; BDA sources these two screens from GeneDisco, whose benchmark explicitly uses the *interference* screens (Mehrjou et al., 2021, §4.2.2–4.2.3), although BDA's own task prompt does not state the arm. This matters here: it makes all eight non-Steinhart screens loss-of-function, leaving Steinhart the lone gain-of-function screen (Table 7). ‡The two Perturb-seq screens define a hit as a strong *transcriptome-wide* response (top 10%), not a specific phenotype at the top 5% used for the other seven — which is why their hit rates are higher.

Steinhart is not a published study: it is unpublished data generated by a BDA co-author (Z. Steinhart, Marson lab), which BDA included specifically as a leakage control — "one dataset that is unpublished, ensuring it is not part of the language model's training data". That control was valid for *their* models but **not for ours**: BDA released the screen in their public repository in 2024, before our LLM's training cutoff. See Limitations.

			all → non-essential			
Dataset	Task (BDA prompt)	Perturb.	Genes	Hits	Hit rate	Batch
IFN-γ	IFN-γ production*	CRISPRi	18,418 → 17,785	920 → 802	4.5%	128
IL-2	IL-2 production*	CRISPRi	18,939 → 18,273	654 → 481	2.6%	128
Sanchez21	Tau change, *either* direction	KO	18,469 → 17,807	924 → 859	4.8%	128
Sanchez21 (dn)	Tau *decrease*	KO	18,469 → 17,807	924 → 871	4.9%	128
Carnevale22	T-cell efficacy (adenosine)	KO	18,861 → 18,224	943 → 868	4.8%	128
Scharenberg22	Lysosomal choline recycling	KO	1,061 → 1,029	53 → 49	4.8%	32
Steinhart	**Resisting T-cell exhaustion**	CRISPRa	18,797 → 18,144	149 → 145	0.8%	128
K562-Essential	Strong transcriptomic response‡	CRISPRi	623 → 623	63 → 63	10.1%	32
K562-GWPS	Strong transcriptomic response‡	CRISPRi	9,193 → 9,193	924 → 924	10.1%	128

Relationship to BioDiscoveryAgent

We build on BDA's *data*: the ground-truth phenotype scores and `topmovers` hit sets are theirs, and we verified that our hit labels are exactly `topmovers` ∩ (pool \ CEGv2) for every screen. Our headline numbers are not a reproduction of theirs — we add components they do not have — but to make the comparison controlled rather than transcribed, **we did run BioDiscoveryAgent's own agent on our benchmark** (Table 2, and below). Two differences then explain our lead:

- **We add a cross-experiment ML prior that BDA does not have.** Our LOO-LightGBM is trained on the *other* screens' hit labels using PPI / KEGG / pLI / DepMap features. Restricted to the six screens we share with BDA, that model alone averages 0.187 (Table 2) — already well above BDA's best agent (0.127) and the best GeneDisco baseline (0.091), *with no LLM at all.* Our gains over BDA's published numbers should therefore be read as "a cross-experiment prior helps a lot", not as "our agent is a better agent".

- **Different LLM and different baseline implementations** (we use Claude Haiku 4.5; BDA's headline model is Claude 3.5 Sonnet; our Coreset is our own implementation and scores differently from theirs, e.g. 0.100 vs. 0.066 on IFN-γ).

To go beyond published numbers we ran **both** prior methods on our benchmark. BioDiscoveryAgent's own no-tools agent, with the same LLM as our C-arm (Claude Haiku 4.5) and scored by our oracle, reaches **0.105** — close to the 0.127 Roohani et al. report for their Sonnet 3.5 agent, confirming the reference is fair. GeneDisco is a benchmark of active-learning *acquisition functions* (a surrogate retrained on the genes tested so far, plus a rule that picks the next batch), not a model, so we ported its acquisition rules verbatim from source onto *our* feature table and oracle — giving its methods our features and isolating the acquisition strategy, faithful to its essence that the surrogate sees only within-experiment labels (round 1, with no data, falls back to random, as its own loop does). Its best-per-screen reaches **0.113**, close to the transcribed 0.091.[1] As a convention check, our random baseline agrees with theirs on some screens (IL-2 0.031 vs. 0.031) but differs by up to 0.034 on others, so treat transcribed gaps below $\sim$0.03 as noise.

Table 2: The six screens BDA and we share (Schmidt1/2 = IFN-γ/IL-2, CAR-T = Steinhart, plus Scharenberg22, Carnevale22, Sanchez21). Mean hit@R5 over the non-essential (N/E) evaluation, 5 rounds, same batch sizes. The two **controlled** rows are *our runs* on our benchmark: BioDiscoveryAgent's agent (Haiku 4.5, our eval)* and GeneDisco's acquisition functions ported onto our feature table. The transcribed rows are from Roohani et al. (2025), Table 1. Both "best-per-screen" entries are per-screen oracles over the acquisition rules considered — the transcribed one over their *eight* baselines, our controlled one over the *five* rules we ported (uncertainty top / soft, margin, coreset, k-means-data; we omit BADGE, k-means-embedding, and DiscoBax, which need model internals we do not reproduce). The controlled oracle is thus over fewer rules, if anything understating it.

*We route BDA's LLM calls through our client (for auth) and use our model; unused tool dependencies are stubbed, the output parser is relaxed to tolerate a missing "Solution:" header, and a harness bug that discarded an unfilled final round is fixed. None of these touch the agent's reasoning or gene choices.

Method	Source	mean hit@R5
Random	BDA (transcribed)	0.049
Random	ours	0.041
GeneDisco — best-per-screen (oracle over their 8)	BDA (transcribed)	0.091
GeneDisco — best-per-screen (oracle over our 5), controlled	**ours (run)**	0.113
BioDiscoveryAgent — Claude 3.5 Sonnet	BDA (transcribed)	0.127
BioDiscoveryAgent — controlled	**ours (run)**	0.105
LOO-LightGBM — **no LLM** (ours)	ours	**0.187**
Waddington C (ours)	ours	**0.217**

[1] Running GeneDisco's *vanilla package* on our screens instead gives near-random results ($\sim$0.03), but that is an artefact of its weak native features (Achilles): there its surrogate has $\approx$0 rank correlation with the phenotype, which collapses every acquisition rule to random. Given our features the same rules work, so this reflects the feature set, not the algorithms.

The *ordering* is the point. Run controlled on our benchmark, both prior methods land in the same place — GeneDisco's active learning at 0.113, BioDiscoveryAgent's LLM agent at 0.105. **Our cross-experiment prior alone — a static LightGBM with no LLM (0.187) — nearly doubles either.** The gap is not acquisition cleverness or LLM quality: GeneDisco and BDA both learn only *within the single experiment*, whereas our LOO transfers the other screens' hit labels. The honest reading is a hierarchy of *information*, all on these same six screens — learning inside one screen (GeneDisco active learning ≈ BDA agent ≈ 0.11) < a cross-experiment supervised prior (LOO 0.187) < that prior plus online adaptation and LLM verification (C, 0.217 here; 0.253 over all nine screens, Table 4) — not "our agent beats their agent". Two per-screen notes, both consistent with the rest of the paper: controlled BDA does its best relative to random on Steinhart (0.117, above even our LOO's 0.076), the CRISPRa screen where the ML prior's knockout-derived features have least to say; and GeneDisco's within-experiment active learning is *not* weak everywhere — on the high-hit-rate K562-Essential (outside the shared six) it reaches 0.651, above even our C — it fails only where the hit rate is low and a prior matters most.

Table 3: Per-screen breakdown behind Table 2. **BDA-ctrl** is our controlled run (Haiku 4.5, our eval); **BDA-trans** is transcribed (Sonnet 3.5). Our controlled BDA tracks the transcribed number closely on four of six screens (validating the reference) and falls below it on Scharenberg22 and Sanchez21 (where Haiku trails Sonnet). Our C beats the controlled agent on every screen; our LOO — with no LLM — beats it on five, the exception being Steinhart.

Screen	Random	BDA-ctrl	BDA-trans	LOO (no LLM)	**C (ours)**
IFN-γ	0.029	0.097	0.107	0.168	**0.190**
IL-2	0.031	0.118	0.122	0.306	**0.347**
Steinhart	0.021	0.117	0.133	0.076	**0.159**
Scharenberg22	0.102	0.225	0.292	0.449	**0.465**
Carnevale22	0.024	0.045	0.044	0.048	**0.056**
Sanchez21	0.037	0.027	0.063	0.077	**0.087**
Average	0.041	0.105	0.127	0.187	**0.217**

What the cross-experiment prior actually learns (and what it does not)

The prior is the largest single source of our advantage, so it deserves scrutiny: is its gain genuine phenotype-specific transfer, or something more trivial — sibling-screen leakage, generic essentiality, plain hit recurrence? We probe this with the LOO LightGBM alone (no LLM), on a uniform all-feature baseline so the feature-family ablation is clean; this scores 0.243 averaged over the nine screens (slightly above the routed 0.187/0.217 used elsewhere, as it keeps DepMap on every screen). Three probes, and the picture is sobering.

- **Sibling leakage is real.** Our nine screens contain three same-study pairs (IFN-γ/IL-2, Sanchez21/-down, K562-Essential/-GWPS). Re-training with the target's sibling *also* removed drops the prior from 0.243 to 0.199 on average, and from 0.248 to **0.182** on the six paired screens — concentrated on IL-2 (-0.098) and K562-Essential (-0.222), whose Perturb-seq sibling shares much of its hit structure. So "generalises to a new screen" partly means "transfers from a sister screen in the same study".
- **Most of the signal is generic essentiality.** Restricted to the DepMap features alone the prior scores 0.230 — nearly the full 0.243 — while the phenotype-specific features (intrinsic

+ anchor, *no* DepMap) give 0.219. The prior leans heavily on knockout essentiality, which is not screen-specific.

- **A trivial baseline matches it.** Ranking a gene simply by *how often it is a hit in the other screens* — no model, no features — also averages **0.243**. The LightGBM adds nothing over counting cross-screen hit recurrence.
- **The anchor features leak, but are not load-bearing.** The three anchor-relative features (proximity to a phenotype's seed genes) are built from a hand-picked anchor set that, by our own audit, is **51% composed of the screen's own hits** (42 of 82 anchors; e.g. 10/12 for IL-2, 9/10 for K562-GWPS) — genuine privileged information. Yet dropping *every* anchor-relative feature costs the prior only 0.015 ($0.243 \to 0.228$): the anchor family is the weakest on its own (0.156), and the label-free DepMap (0.230) and intrinsic-topology (0.218) features are what carry it. The leak is real; its contribution to the headline is not.

A novel-hit analysis explains the mechanism: the prior's gains sit on screens whose hits recur (K562-Essential has only 6% novel hits, IL-2 41%), and it is near random on screens dominated by novel, phenotype-specific hits (Carnevale22 80% novel, Sanchez21 72%). **The cross-experiment prior succeeds by exploiting recurrence and essentiality, not by discovering new phenotype-specific biology.**

This qualifies but does not overturn the headline. Across their respective evaluation sets, all three corrections — siblings excluded (0.182), DepMap removed (0.219), anchor features removed (0.228) — leave a substantial transferable signal; because these are measured on different screen subsets, we do not read them as a single numeric hierarchy against the ~ 0.11 single-experiment methods (Table 2). The honest claim is therefore narrower and, we think, more useful: *a screen's hits are substantially predictable from other screens' labels, largely through recurrence and essentiality* — which no within-experiment active learner or LLM agent can see — but this prior is not a discovery engine for genuinely new biology. We tested directly whether the within-screen components add value *over* the raw recurrence count: starting both from the same single hit-frequency feature (5 seeds), adding online retraining and LLM verification moves the mean from 0.243 to 0.239 (paired −0.004, screen-clustered 95% CI $[-0.046, 0.028]$; higher on 6 of 9 screens but lower on average) — **indistinguishable from, if anything slightly below**, the pure recurrence ranking. So online adaptation and LLM verification do lift the *feature-based* prior — against the matched final-configuration LOO, $0.231 \to 0.253$ (+0.022, CI $[0.012, 0.036]$) — but we cannot claim they beat a pure recurrence ranking. All numbers here are produced by `python -m waddington_select.analysis.prior_probes` and `.router_analysis`.

Recurrent vs. novel hits (exploratory). A sharper question than mean hit@R5: can any method find hits that *never* occur in the training screens — biology recurrence cannot see? Splitting each screen's hits into recurrent (a hit in ≥ 1 other screen) and novel (never), 65% of all hits are novel. A hit-frequency ranking found **none** of them in these runs (novel-recall 0.000, as every novel gene receives a zero recurrence score) while recovering 49.5% of recurrent hits. The *feature-based* LOO prior does better on novelty (macro novel-recall **0.105**, recurrent 0.439): its structural features flag a plausible hit even when the gene has never been a hit elsewhere. The full Waddington does *not* improve on this — novel-recall 0.074 (lower on 6 of 9 screens; paired −0.031, screen-clustered 95% CI $[-0.089, 0.007]$) while recurrent-recall rises to 0.462 (+0.023, $[-0.012, 0.054]$). The *point estimates suggest* that the *combined* within-screen components (online retraining + LLM endorsement) shift selection toward recurrent, high-confidence hits and away from novel ones, **although neither paired interval excludes zero**; the comparison also does not isolate whether the online ML or the LLM drives it. This pattern is *consistent with, but does not independently establish*, the interpretation of the LLM as a verifier of confident candidates rather than a discoverer of new biology. We report it as exploratory (no tuning on it); `python -m waddington_select.analysis.novel_hit_analysis` and `.final_system_stats`.

Table 4: Hit ratio at round 5 across all nine benchmark datasets. All six methods are reported over 20 seeds. **Bold** = best per row; underline = second best.

‡**Column C is the final leakage-free Waddington** (non-target-selected weight, no anchor features; 20 seeds; §), avg 0.253. **The LOO-LightGBM and OnlineAdapt columns are benchmark baselines under a *different* feature configuration**; the within-system *matched* prior (same no-anchor metadata features) is 0.231, so the internal improvement is 0.231 → 0.253 (+0.022, Table 5), not 0.217 → 0.253. The detailed ablations, attribution, and SHAP that follow were computed on the legacy target-aware routed configuration (avg 0.256, no detected difference in aggregate; per-screen values in the reason-vs-recall table) and are kept as supporting analyses of that near-performing predecessor. Relative to the legacy variant the final system trades a little on the gain-of-function screen Steinhart (0.124 vs. 0.159, where the LLM weight and gene-name signal matter most) for gains on Scharenberg22 and K562-Essential.

†**B is not a pure-LLM baseline.** When the LLM names too few genes that exist in the pool, the batch is padded from the static LOO-LightGBM ranking. Averaged over the five rounds that padding is 19% (IFN-γ), 19% (IL-2), 46% (Sanchez21), 32% (Sanchez21-dn), 7% (Carnevale22), 68% (Scharenberg22), 43% (Steinhart), **86% (K562-Essential)** and 0% (K562-GWPS). B should be read as "LLM, padded with a static ML prior", and its two strongest entries (K562-Essential 0.544, Scharenberg22 0.466) are largely that prior rather than the LLM.

Dataset	Random	LOO-LightGBM	OnlineAdapt.	A: Coreset	B: LLM†	C: Waddington (final)‡
IFN-γ	0.029	0.168	**0.183**	0.100	0.159	<u>0.176</u>
IL-2	0.031	0.306	<u>0.314</u>	0.139	0.259	**0.326**
Sanchez21	0.037	0.077	**0.087**	0.031	0.064	<u>0.086</u>
Sanchez21 (dn)	0.029	<u>0.091</u>	**0.101**	0.078	0.068	0.091
Carnevale22	0.024	0.048	0.047	0.054	<u>0.057</u>	**0.069**
Scharenberg22	0.102	0.449	0.449	0.286	<u>0.466</u>	**0.510**
Steinhart	0.021	0.076	0.090	0.090	**0.157**	<u>0.124</u>
K562-Essential	0.254	0.492	0.476	0.270	<u>0.544</u>	**0.571**
K562-GWPS	0.065	0.247	<u>0.273</u>	0.156	0.221	**0.321**
Average	0.066	0.217	<u>0.224</u>	0.134	0.222	**0.253**

The final Waddington achieves the best average hit ratio (0.253), above the padded LLM baseline B (+0.031) and the diversity baseline A (+0.119). It is best on five screens (IL-2, Carnevale22, Scharenberg22, and both K562 screens) and a close second on three (IFN-γ, Sanchez21, Steinhart). On Sanchez21-dn it ties the LOO prior for second, behind OnlineAdaptive by 0.010. Its one clear give-up is **Steinhart**, where it trails the LLM baseline by 0.033: that is the gain-of-function (CRISPRa) screen where gene-name semantics and a higher LLM weight help most, and the leakage-free global w_{LLM}=0.2 forgoes both (the legacy routed variant scored 0.159 there). Even that baseline is 43% static-ML padding (Table 4†), so its advantage cannot be cleanly attributed to the LLM alone; the same padding caveat applies to B's high marks on Scharenberg22 and K562-Essential, which the final system nonetheless beats outright.

Robustness: cluster on the screen, not the gene. The nine screens are wildly heterogeneous (hit@R5 ranges from 0.06 on Steinhart to 0.55 on K562-Essential), so a bootstrap that treats the *screen* as the resampling unit gives deliberately wide marginal intervals: C 0.256 $[0.151, 0.372]$ (this robustness analysis is on the legacy routed C; the final system's 0.253 sits inside the interval), the LOO prior 0.217 $[0.120, 0.323]$, Online ML 0.224 $[0.128, 0.324]$ — all overlapping. On an unpaired basis, with nine screens, no method is separable from another, and we do not claim otherwise. The comparison that survives is the *paired* one, where the between-screen variance cancels:

Final Waddington − baseline (paired, 9 screens)	mean Δ	95% CI (screen-clustered)	win/tie/loss
vs. LOO prior	+0.036	$[+0.017, +0.055]$	8 / 1 / 0
vs. Online ML	+0.028	$[+0.008, +0.052]$	6 / 0 / 3
vs. LLM branch	+0.031	$[+0.009, +0.054]$	8 / 0 / 1
vs. Coreset	+0.119	$[+0.058, +0.187]$	**9 / 0 / 0**

These are the *final* 20-seed leakage-free system's paired deltas (not the legacy C). All paired mean deltas are positive and their intervals exclude zero; Waddington wins on 6–9 of the nine screens, depending on the comparator (6/0/3 vs. Online ML, 9/0/0 vs. Coreset). So the defensible statement is not "its mean is significantly higher" — with nine heterogeneous screens it is not — but "it improves on each baseline consistently on the majority of screens." Reproduce with `python -m waddington_select.analysis.final_system_stats`.

Robustness to a leakage-free router

The Method disclosed a real hazard: the shipped router reads the target's *realized* hit rate to pick the fusion weight (Method, "What the router may and may not know"). We now test directly whether that leakage carries the result. Under a protocol **frozen and committed before these numbers were inspected** (`waddington_select.router_protocol`), we re-derive the fusion policy using only the *other* screens and metadata knowable *before* the target screen runs (pool size, perturbation modality, cell line) — never the target hit rate. Two variants: **global-fixed**, one fusion weight w_{LLM} chosen by leave-one-out over the other screens; and a **nested router**, $w_{\mathrm{LLM}} = f(n_{\mathrm{pool}})$ with thresholds tuned only on the other screens. Both use a label-free metadata feature policy (DepMap dropped iff CRISPRa or curated-essentiality; cell-line-matched DepMap dropped on its own cell line). We report leave-one-study-out at two strengths, kept distinct: one removes the same-study group only from the router's *config selection*; the *full* version additionally denies the cross-experiment *prior* its siblings.

Table 5: **The final Waddington is the leakage-free, privilege-free configuration** (bold): its fusion weight is selected on non-target screens, its feature policy is a pure function of pre-experiment metadata, and it uses *no* anchor-relative features. It reaches 0.253 — with no detected difference from the legacy target-aware routed C-arm (0.259, which reads the realized hit rate). The within-system gain over its *matched* static prior (same no-anchor metadata features, 0.231) is +0.022; against the differently-configured external LOO baseline (0.217) it is +0.036. Deltas are paired per-screen with screen-clustered bootstrap 95% CIs (nine screens). The variant rows show what each relaxation is worth (anchor features +0.009; per-screen routing 0); the last row is the strict test that also denies the prior its siblings. Reproduce with `python -m waddington_select.analysis.router_analysis` and `.final_system_stats`.

Configuration	mean	Δ vs. stated ref. (W/T/L)	Δ vs. legacy C (W/T/L)
Differently-configured LOO (external)	0.217	—	
Matched final-config prior (no-anchor)	**0.231**	—	
Legacy routed C (*uses realized hr*)	0.259	+0.042 [.020, .066] 9/0/0	—
Final Waddington (no anchor feats)	**0.253**	+0.022* [.012, .036] 8/0/1	−0.006 [−.019, .007] 3/1/5
with anchor features (global-fixed)§	0.264	+0.047 [.025, .071] 9/0/0	+0.008 [−.013, .031] 6/0/3
nested router (with anchor)	0.260	+0.043 [.021, .069] 9/0/0	+0.004 [−.017, .029] 5/0/4
study-excluded policy selection	0.262	+0.045 [.021, .070] 8/0/1	+0.006 [−.015, .030] 4/0/5
Full leave-one-study-out (policy *and* prior)	0.239	+0.041† [.014, .078] 7/0/2	—

*The Final Waddington Δ is against its *matched* static prior (0.231); against the differently-configured external LOO baseline (0.217) it is +0.036 [.017, .055], 8/1/0. The variant rows below are against the 0.217 baseline. §Final Waddington and the legacy routed C are over 20 seeds (matching Table 4); the router-variant rows are 3 seeds, so the anchor / routing deltas are read at 3 seeds (0.264 vs. 0.255). †For the last row the prior is also denied its siblings, so the matched reference is the *sibling-excluded* LOO (0.199), not the standard LOO; against the standard LOO (0.217) it is +0.022 [−.004, .052]. "+ anchor" / "nested" / "study-excluded" rows keep siblings in the prior. Ablations, attribution, and SHAP elsewhere in the paper were computed on the legacy routed C-arm (at 5 seeds, avg 0.256; no detected difference from the final system in aggregate) and kept as supporting analyses; component effects were not re-estimated under the final configuration.

This is the configuration we report as Waddington. It fixes $w_{\mathrm{LLM}} = 0.2$ globally — selected by leave-one-out over the *other* screens, unanimously in every fold — uses the metadata feature policy, and drops the anchor-relative features flagged as privileged. Over 20 seeds it reaches 0.253 — with no detected difference from the legacy target-aware routed C-arm (0.259; paired −0.006, CI [−.019, .007]) — while using *no* target statistic and *no* privileged feature. Over its *matched* static prior (0.231) the within-system gain is +0.022 (95% CI [0.012, 0.036], 8/0/1); against the differently-configured external LOO baseline (0.217) it is +0.036 [0.017, 0.055]. The two relaxations the legacy system made buy almost nothing: at 3 seeds restoring the anchor features adds only ≈0.009, and the hit-rate per-screen routing adds nothing at all — it was in fact mildly *suboptimal*, forcing $w_{\mathrm{LLM}} = 0.4$ on the two small screens where a global 0.2 scores higher (Scharenberg22 0.55 vs. 0.45; K562-Essential 0.59 vs. 0.55). We keep the legacy routed numbers for the detailed ablations, attribution, and SHAP that were computed on it (no detected difference in aggregate) as supporting analyses of that near-performing predecessor, without assuming every component effect transfers quantitatively; the headline rests on the leakage-free system.

Robustness to seed count. The reported numbers use 20 seeds. Re-running the six main-table arms at 5 seeds instead leaves the four deterministic baselines identical to three decimals (Random 0.066, Coreset 0.134, LOO-LightGBM 0.217, OnlineAdapt 0.224); only the two LLM-driven arms

Table 6: **Per-screen hit@R5 at 5 seeds**, a robustness companion to the 20-seed Table 4. The four deterministic arms (Random / LOO / OnlineAdapt / Coreset) are identical to the 20-seed table; only the two LLM-driven arms (B, C) move, within run-to-run noise.

Screen	Random	LOO-LightGBM	OnlineAdapt.	A: Coreset	B: LLM	C: Waddington
IFN-γ	0.029	0.168	0.183	0.100	0.160	0.183
IL-2	0.031	0.306	0.314	0.139	0.257	0.343
Sanchez21	0.037	0.077	0.087	0.031	0.063	0.078
Sanchez21 (dn)	0.029	0.091	0.101	0.078	0.071	0.088
Carnevale22	0.024	0.048	0.047	0.054	0.059	0.065
Scharenberg22	0.102	0.449	0.449	0.286	0.478	0.490
Steinhart	0.021	0.076	0.090	0.090	0.152	0.117
K562-Essential	0.254	0.492	0.476	0.270	0.559	0.571
K562-GWPS	0.065	0.247	0.273	0.156	0.225	0.324
Average	0.066	0.217	0.224	0.134	0.225	**0.251**

move, within run-to-run noise (Waddington 0.253 → 0.251, LLM branch 0.222 → 0.225). The within-system gain over the matched static prior is then +0.020 (95% CI $[0.007, 0.035]$, 7/0/2), marginally wider than the 20-seed estimate (+0.022, $[0.012, 0.036]$, 8/0/1) but same-signed, with every secondary comparison still interval-excluded from zero. The headline is therefore stable to the seed count; per-screen 5-seed values are in Table 6.

The stricter test: deny the prior its siblings. The variants above route honestly but still let the cross-experiment prior *train on* same-study sibling screens. The strongest leave-one-study-out also removes the sibling from the prior itself. This does bite: the mean falls from 0.264 to **0.239**, almost entirely on two screens that lose an informative sibling — K562-Essential 0.587 → 0.444 (its Perturb-seq twin) and IL-2 0.358 → 0.308 (IFN-γ). But this handicaps only the numerator. The identical removal drops the bare LOO prior in lockstep (0.243 → 0.199; the sibling-excluded prior of the "what the prior learns" analysis). Compared *like-for-like* — both the full system and the prior denied their siblings — the C-arm still beats the prior by +**0.041** (screen-clustered 95% CI $[0.014, 0.078]$, winning 7 of 9). So the same-study structure inflates the *absolute* hit rates but not the C-over-prior increment that is the actual claim: that increment comes from within-screen online adaptation and LLM verification, which no sibling supplies.

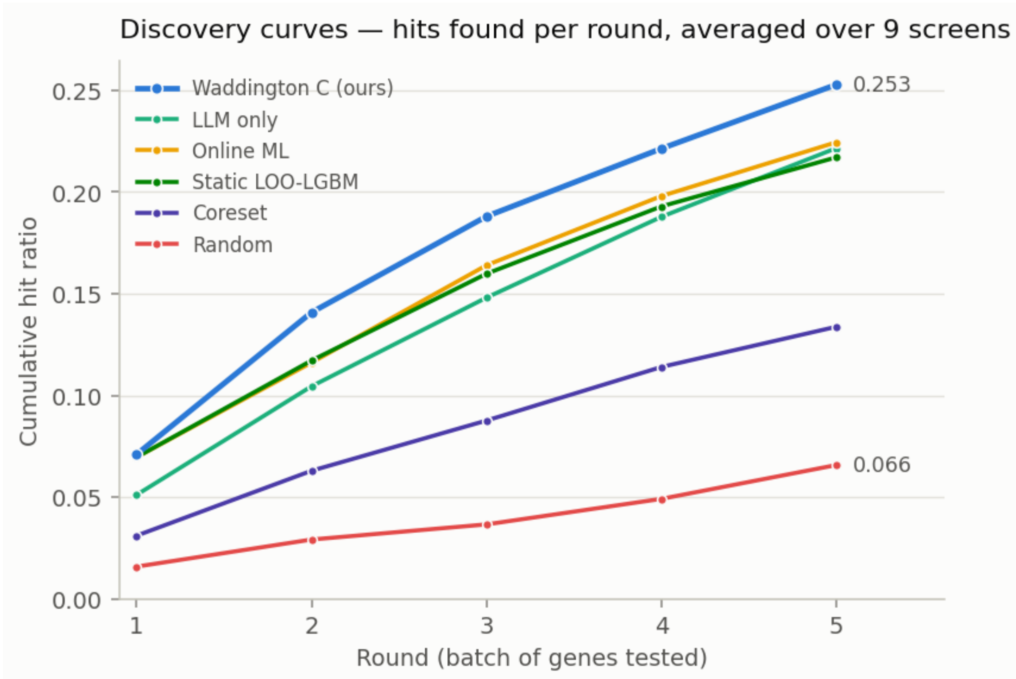

Figure 1: Discovery curves: cumulative hit ratio per round, averaged over the nine screens, for the **final leakage-free Waddington** (20 seeds, hit@R5 0.253). The hybrid separates from every baseline from round 2 onwards. Per-screen values are given in Figure 3; no error band is drawn because the spread across screens is dominated by how hard each screen is (K562-Essential 0.56 vs. Carnevale22 0.06), which would encode screen difficulty rather than uncertainty about a method.

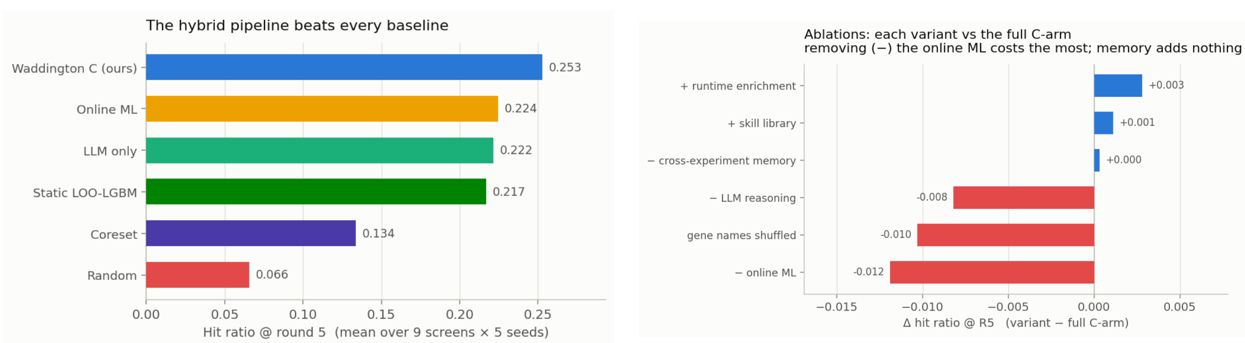

Figure 2: Left: average hit ratio @ R5 per method, final leakage-free Waddington 0.253 (Table 4). Right: paired ablation deltas (Table 11) — a *legacy configuration analysis* (variants share the legacy routed C-arm reference, which shows no detected difference from the final system in aggregate). Colour states a fact about the *variant* (it scored lower than the full arm), not a verdict on the component: a red "− online ML" means *removing* online ML costs 0.012, i.e. it is the largest loss in this legacy ablation.

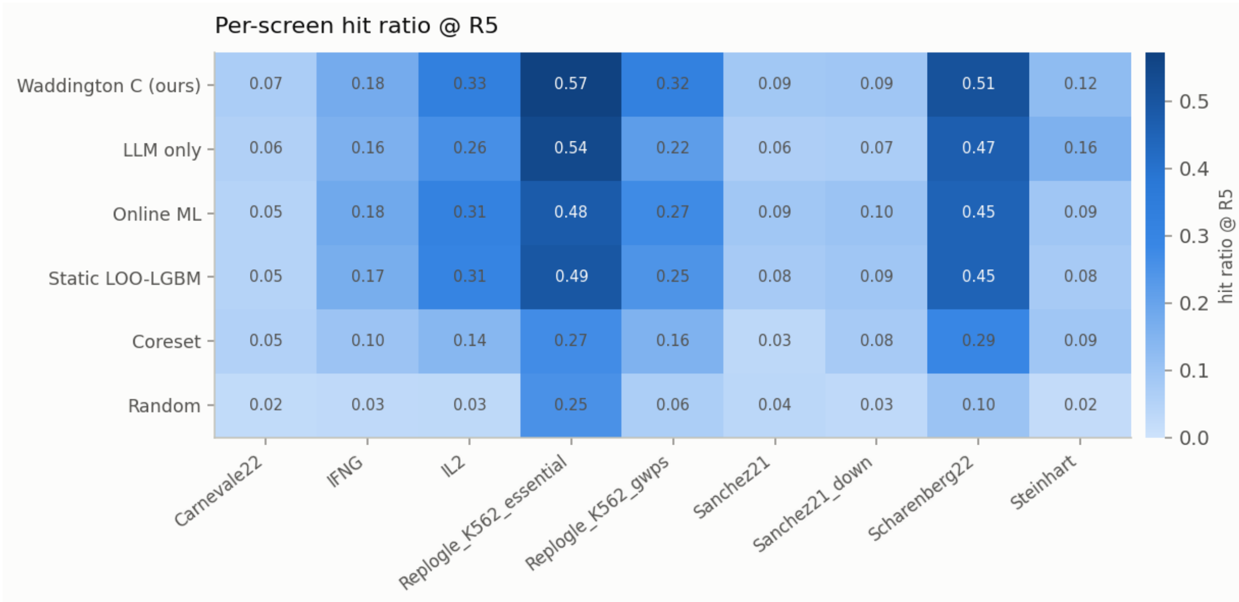

Figure 3: Per-screen hit ratio at round 5 for every method.

In the legacy predecessor, removing online retraining produced the largest average loss, while the LLM-branch ablation was strongly confounded by static-prior padding. Gene-name shuffling reduced average performance by 0.010 (and by 0.084 on Steinhart), motivating the controlled structure-versus-name experiment below; the full stepwise decomposition and per-screen ablation tables are in the Appendix (Tables 10–11).

Reasoning vs. recall: can structure replace the gene name?

The shuffle ablation shows the LLM's contribution runs through gene-name knowledge, but it does not show *why* — because it removes the name and supplies nothing in its place. The shuffled arm must name identifiers out of an 18k space with no information whatsoever; it measures a blindfolded guesser, not a reasoner. The question we actually care about is whether the LLM must *recall* biology from its weights, or whether it can *reason* from structure we hand it.

We therefore ran a four-point decomposition. All four use the identical routing, fusion and update harness; the *only* thing that varies is what the LLM sees:

- **A0** — anonymized IDs, no candidate pool, no features (the shuffle arm above).
- **A0.5** — anonymized IDs + the ML-ranked candidate pool, *no* features (control).
- **A1** — anonymized IDs + the pool + each candidate's structural feature vector, plus, each round, the mean feature profile of the revealed hits vs. non-hits, so the LLM can induce a decision rule and apply it.
- **A2** — real gene names (the C-arm).

Two design points matter. A1 is never shown the ML confidence score: were it shown, "reasoning" would collapse into copying the GBM. And **A0.5 exists because A1 is handed an ML-ranked pool that A0 is not** — without that control, A1 − A0 would confound "the features" with "being given a shortlist", and on the two smallest screens (K562-Essential: 623 genes; Scharenberg22: 1029) that shortlist is itself a strong prior.

Table 7: Reasoning vs. recall. Hit ratio at round 5, mean over **5 seeds**, same backend, $T = 0$. The three right-hand columns isolate one ingredient each, holding the others fixed, and are *seed-paired* (mean of per-seed differences) ± 1 s.d. — not differences of means. A2 reproduces its committed 5-seed values on all nine screens (max drift 0.025).

	hit@R5 (mean)					value of … (seed-paired ±s.d.)		
Screen	A0	A0.5	A1	A2	GBM	pool A0.5−A0	**features** A1−A0.5	name A2−A1
IFN-γ	0.180	0.160	0.175	0.190	0.183	−0.020 ±.003	+0.015 ±.008	+0.015 ±.015
IL-2	0.351	0.316	0.343	0.347	0.314	−0.035 ±.000	+0.027 ±.007	+0.004 ±.004
Sanchez21	0.088	0.090	0.090	0.095	0.087	+0.002 ±.002	+0.001 ±.005	+0.004 ±.005
Sanchez21 (dn)	0.091	0.093	0.099	0.095	0.101	+0.002 ±.002	+0.006 ±.001	−0.003 ±.005
Carnevale22	0.064	0.051	0.058	0.059	0.047	−0.013 ±.001	+0.007 ±.002	+0.001 ±.002
Scharenberg22	0.469	0.531	0.584	0.449	0.449	+0.061 ±.000	+0.053 ±.011	**−0.135** ±.011
Steinhart	0.075	0.048	0.065	0.145	0.090	−0.026 ±.003	+0.017 ±.008	**+0.080** ±.013
K562-Essential	0.286	0.318	0.597	0.549	0.476	+0.032 ±.043	**+0.279** ±.014	−0.048 ±.019
K562-GWPS	0.343	0.300	0.372	0.347	0.273	−0.043 ±.000	+0.072 ±.006	−0.024 ±.006
Average	0.216	0.212	**0.265**	0.253	0.224	−0.005	**+0.053**	−0.012

Three results, in order of how much they surprised us:

1. **The shortlist is worth nothing** (−0.005, negative on 5 of 9). A1's gain is not an artefact of being handed ML-curated candidates.
2. **The features are worth** +0.053**, and positive on all nine screens** (+0.001 to +0.279), every per-screen effect exceeding its seed s.d. — the most consistent effect anywhere in this paper.
3. **The gene name is worth** −0.012**.** With every gene renamed to a meaningless identifier and the structural features supplied, the system *matches or slightly exceeds* the real-name arm (0.265 vs. 0.253). On average, *the LLM does not need to know that a gene is IRF4; it needs the structure.*

Is this the LLM *reasoning*, or would any rule do? A1 hands the LLM the feature vectors and lets it induce a decision boundary. To test whether the *LLM* is what matters, we replace it with the most trivial boundary imaginable and change nothing else: score each candidate by its projection onto the (hit-centroid − non-hit-centroid) axis in the identical anonymized feature space, take the top B, no language model (round 1 falls back to the static ranking, exactly as A1 pads). This one-line rule scores **0.268** averaged over the nine screens — a *slightly higher mean* than A1's 0.265, with mixed per-screen differences (K562-Essential 0.587 vs. 0.597, Scharenberg22 0.571 vs. 0.584, Steinhart 0.083 vs. 0.065, IL-2 0.351 vs. 0.343). So A1 is **not** evidence that the LLM reasons: the value is that the structural features are linearly separable, and a centroid rule extracts it as well as a language model does. What the decomposition establishes is therefore narrower — and cleaner — than "the LLM reasons": the result is *consistent with gene-name dispensability on average under this feature representation*, with the LLM merely one (here interchangeable, and strictly unnecessary) way to map that structure to a choice.

Do these features smuggle in privileged information? The structural features include the anchor-relative ones the leakage audit flagged as privileged (51% of anchors are themselves target hits). If "structure replaces names" rode on that leak, it should collapse once the anchor features

are removed. It does not. Re-running the ladder in the anchor-free feature space, structure still gives *similar* performance to names — A1 0.241 vs. A2 0.249 (paired −0.007, 95% CI [−0.041, 0.029]; no detected difference, though the interval does not establish equivalence) — and the linear rule does about as well as the LLM (0.257; +0.008 vs. names). Removing the privileged features lowers both structure arms in step (they were the strongest single family), so the *contrast* is preserved: these results are consistent with gene-name dispensability on average under the tested feature representation, even when the structure the LLM sees contains no target-hit leakage. (Reproduce with `WADDINGTON_DROP_ANCHOR_FEATS=1`.)
That average, however, hides a bimodality that is the whole point:

- **Steinhart: the name is worth** $+0.080 \pm 0.013$, the largest anywhere, and the features recover almost nothing ($+0.017 \pm 0.008$; A1 at 0.065 is even *below* A0 at 0.075, and the two error bars do not come close to overlapping). This is exactly the screen of Table 12: the one gain-of-function (CRISPRa) screen, where every DepMap feature is knockout-derived and does not transfer. The features we are able to hand the LLM are precisely the ones that are uninformative there, so nothing we supply substitutes for knowing what IRF4 does.
- **Scharenberg22: the name is worth** -0.135 ± 0.011 — it actively hurts, consistent with the shuffle *improving* that screen.

So substituting structure for the gene name is achievable and essentially free on eight of nine screens. Steinhart is the clearest exception, where the available structural features do not recover the benefit associated with gene-name semantics. We read this as a limit of what our feature table *contains*, not as proof that structure cannot suffice: our features are anchor-relative PPI / co-expression / pathway overlap plus constraint and essentiality, and none of them encodes the specific T-cell biology Steinhart rewards. The obvious reading of the bullet above — that Steinhart is special *because* it is the gain-of-function screen — is tempting, and wrong: we tested it directly (next), and perturbation modality does not explain it.

One further calibration: **A0 (0.216) scores below the plain GBM (0.224)**. An LLM branch with nothing to say does not merely fail to help — at a 0.4 fusion weight it drags the hybrid below the ML model it was meant to improve.
Three supporting analyses are deferred to the Appendix: a test of whether the gain-of-function *modality* explains Steinhart's name-dependence (it does not), the SHAP feature attributions of the ML score, and a freely-planning tool-using *agent* that underperforms the fixed-fusion pipeline.

Explainability: what is the LLM actually contributing?

The ablations say *how much* each component is worth; they cannot say *where* the value comes from. We therefore instrumented the arm to record, for every selected gene, a counterfactual attribution against the ML model's own top-k:

- **both** — the LLM named it *and* the ML would have selected it anyway;
- **LLM-only** — the LLM named it and the ML would *not* have selected it (this is the LLM's marginal contribution);
- **ML-only** — the LLM did not name it; it entered on ML score alone.

Only genes the LLM actually named count as LLM picks. The LLM branch pads its batch from the static ranker (Table 4[†]), and crediting that padding to the LLM would make "ML and

LLM agree" partly mean "two ML models agree". An earlier version of this analysis made exactly that error: it reported an agreement bucket of $n = 800$ at a 21.1% hit rate, of which ≈87% was static-ranker padding. Excluding it sharpens the result considerably.

Pairing each pick with its revealed outcome gives the hit rate *per source* (9 screens × 5 rounds, ≈4,200 selected genes). K562-GWPS is reported separately: it is the one screen routed through the two-stage policy (the ML shortlists 384 candidates and the *LLM* makes the final selection), so the "ML-only" category cannot exist there and pooling it would corrupt the rates.

Table 8: Hit rate by attribution source, over the eight weighted-fusion screens. *Pooled* counts every selected gene once; *macro* averages the per-screen rates (screens with ≥ 10 picks in that category). Genuine agreement is rare — only 101 of ≈4,200 picks — but it is by far the strongest signal, at roughly **three times** the ML's own hit rate. The LLM's unilateral picks are *worse* than the ML's when pooled and about equal when macro-averaged, so the defensible claim is that they are *no better*.

Source	Genes selected	Hits	Hit rate (pooled)	Hit rate (macro)
ML + LLM agree	101	47	**46.5%** ± 9.7	**41.5%**
ML only	3,307	487	14.7% ± 1.2	14.9%
LLM only	752	61	8.1% ± 2.0	13.1%

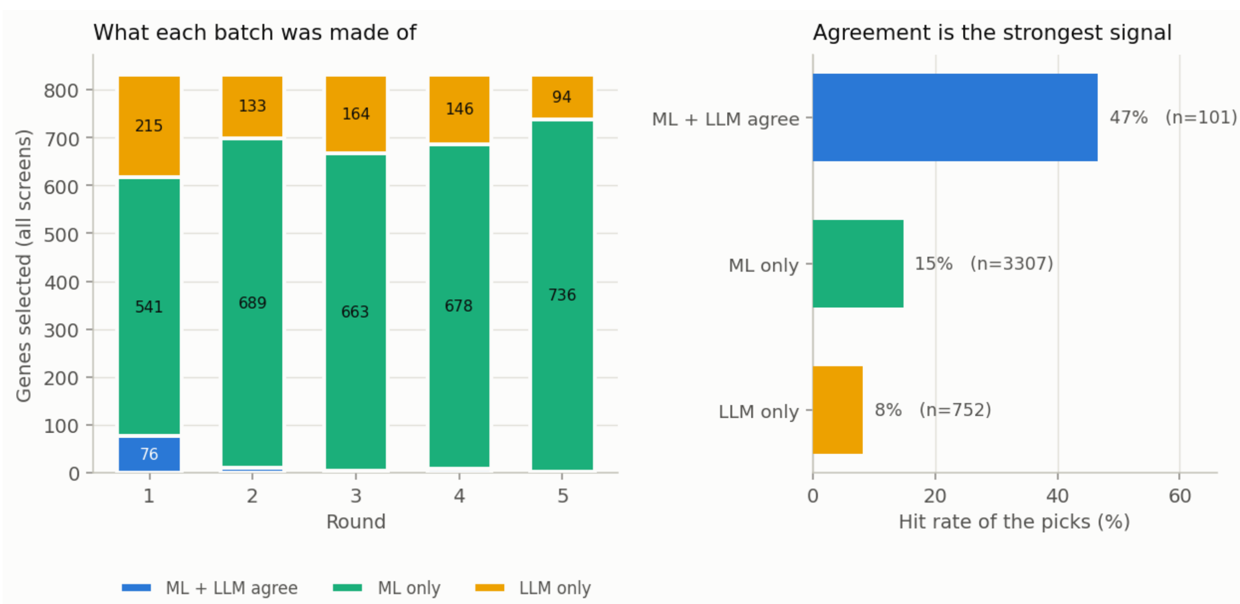

Figure 4: (Legacy configuration analysis.) Left: what each round's batch is made of, pooled over the eight weighted-fusion screens. Right: the hit rate of each source. Genes that *both* components endorse hit at roughly twice the rate of genes the LLM introduces on its own.

The LLM is a verifier, not a proposer. The LLM is a *poor proposer*: left to itself it picks genes that hit at 8.1%, well below the ML's own 14.7%. But it is a *strong verifier*: when it independently names a gene the ML also ranks in its top-k, that gene hits at 46.5% — roughly three times the ML's rate. Such agreement is rare (101 of ≈4,200 picks, i.e. 2.4%), which is precisely why the component is worth so little on average and so much where it fires. This single measurement explains the two results above that otherwise look unrelated:

- why **removing the LLM costs only** 0.008 — the genes it proposes on its own are *worse* than what the ML would have picked instead, so losing them costs nothing; what is lost is a rare but highly informative agreement signal;
- why **giving the LLM tools makes it worse** (−0.047, Table 13) — a free agent selects mostly on its own judgement, which is precisely the weakest of the three sources.

The design implication is the opposite of the prevailing "more agency" direction: the LLM should be used to *re-weight* a calibrated model's candidates, not to choose freely.

Does endorsement add signal *beyond* the ML score? (Yes — but only where the ML is already confident.) The 46.5% agreement rate has an obvious confound: genes both components endorse might simply be the ones with the highest ML score, in which case the LLM adds nothing once that score is held fixed. We test this directly. On the same eight weighted screens, we bin every selected gene into ML-score deciles *within its screen* (so each screen's model scale cancels) and, within each bin, compare the hit rate of genes the LLM named against genes it did not. The value of naming is not constant — it is concentrated entirely in the top of the ML ranking:

Table 9: Hit rate by whether the LLM named the gene, *stratified by within-screen ML-score tertile* (eight weighted screens; the two-stage screen is excluded, its source labels being degenerate). LLM endorsement is worth nothing — slightly negative — in the bottom two tertiles, and worth 2.5× in the top tertile, where the ML is already confident. This is the verifier effect surviving a control for the ML score.

Within-screen ML tertile	LLM named (hit%, n)	not named (hit%, n)	ratio
bottom	7.7% (n=614)	11.3% (n=1050)	0.68×
middle	10.8% (n=130)	15.2% (n=1118)	0.71×
top	**43.1%** (n=109)	17.4% (n=1139)	**2.48**×

A logistic fit on the pooled genes puts a number on it: with the within-screen ML rank percentile and screen fixed effects in the model, LLM endorsement carries an independent odds ratio of **1.68** (screen-clustered bootstrap 95% CI $[1.05, 2.52]$, n = 4,160 genes over eight screens). The interval excludes 1, so endorsement predicts hits *after* adjusting for the ML score — the agreement effect is not merely a restatement of "high ML score." With only eight clusters we also checked leave-one-screen-out: dropping each screen in turn keeps the adjusted OR between **1.39** and **1.87** (all > 1), and removing Steinhart — the natural worry, being the LLM's best screen — leaves it at 1.65, so no single screen drives it. But the tertile split is the sharper statement: the LLM is a verifier of *confident* ML candidates specifically, adding nothing where the ML is unsure and confirming strongly where it is not. That is exactly the regime a fixed-fusion policy exploits and a free agent discards.

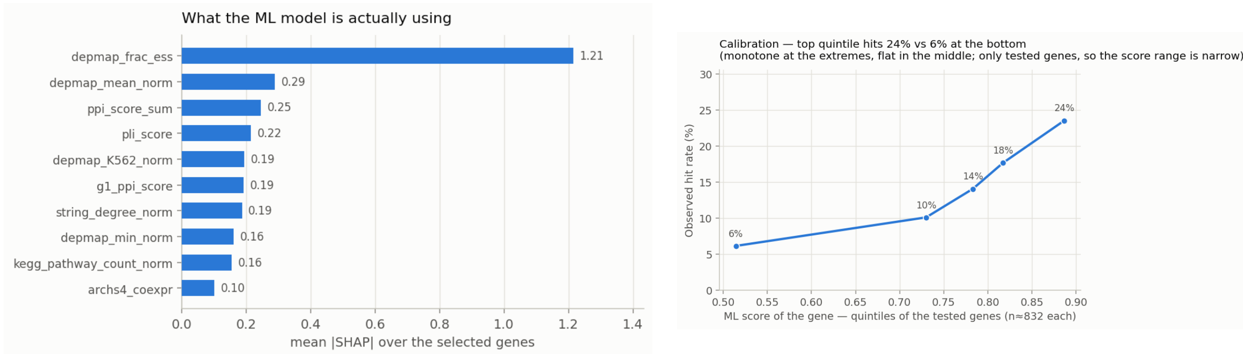

Figure 5: (Legacy configuration analysis.) Left: features driving the ML score over the genes actually selected (LightGBM SHAP contributions; DepMap essentiality and PPI connectivity dominate). Right: calibration — the top-scoring quintile of tested genes hits about twice as often as the bottom quintile, though the curve is flat in the middle and the range is narrow because only *selected* genes are ever tested.

What we claim, and what we do not

Read plainly, the evidence supports a deliberately scoped set of claims:

1. **Cross-screen recurrence and generic essentiality form a strong transferable prior.** A plain hit-frequency ranking over the other screens reaches 0.243, matching the learned LOO-LightGBM; most of the transferable signal is genes that *recur* as hits, not novel phenotype-specific biology.

2. **Waddington improves the feature-based prior it actually deploys.** Against the *matched* final-configuration prior — the static LOO in the same no-anchor metadata feature space (0.231) — the full system (0.253) is +0.022 (screen-clustered 95% CI $[0.012, 0.036]$, 8 of 9 screens); against the paper's differently-configured LOO baseline (0.217) it is +0.036 $[0.017, 0.055]$. Both intervals exclude zero.

3. **We do *not* establish an advantage over a pure recurrence ranking.** Starting both from the same hit-frequency prior, adding online adaptation and LLM verification moves the mean only $0.243 \to 0.239$ (paired -0.004, CI including zero). On this nine-screen benchmark the within-screen components are not shown to beat recurrence.

4. **The LLM's supported role is conditional verification, not discovery.** Endorsement predicts hits after adjusting for the ML score (adjusted OR 1.68), but only among high-confidence ML candidates; it is not shown to surface biology beyond what the ML already ranks highly.

In one sentence: *cross-screen recurrence and essentiality provide most of the transferable signal; Waddington integrates that prior into a constrained, conversational sequential system and reliably improves it through within-screen adaptation and constrained LLM endorsement, but the available nine-screen benchmark does not establish an advantage over a pure recurrence ranking.*

Limitations

We cannot separate the LLM's biological knowledge from memorisation. This is the most important caveat on every LLM result above. Seven of the nine screens are published, so their hit genes may sit in the model's training data, and a gene the model "knows" is a hit is indistinguishable — to our measurements — from a gene it reasons its way to.

It is tempting to appeal to the Steinhart screen as a clean control: BioDiscoveryAgent introduced it precisely because it was *unpublished*, stating that it is therefore "not part of the language model's training data". That argument was sound *for their models* (Claude 3.5 Sonnet, whose training cutoff precedes their release). **It does not transfer to ours.** BDA released the screen in their public repository (`ground_truth_Steinhart_crispra_GD2_D22.csv` and the corresponding `topmovers` file) in 2024, whereas our LLM is Claude Haiku 4.5, released in late 2025. We therefore cannot claim the screen is outside our model's training data, and we withdraw the otherwise attractive inference that "the LLM contributes most on the one screen it provably cannot have memorised".

We judge the practical leakage risk to be low — the hit set exists only as an 18,000-row table of effect sizes and a binary `.npy` array, not the kind of artefact language models reliably memorise — but low is not zero, and the strong claim is not available to us. Note also that the gene-name-shuffle ablation (-0.084 on Steinhart) does *not* settle this: shuffling gene names destroys memorised gene–hit associations and genuine biological knowledge equally, so it cannot distinguish them. Settling the question requires a screen released *after* the model's training cutoff; none of our nine qualifies.

Other limitations.

- **The cross-experiment prior is largely recurrence, not discovery.** As the prior-probe analysis shows, its gain comes substantially from same-study sibling screens (-0.066 on paired screens when excluded) and from generic knockout essentiality (DepMap-only $\approx$ the full model, and a trivial cross-screen hit-frequency count matches it). It is near-random on screens whose hits are mostly novel. It predicts what recurs; it does not find new phenotype-specific biology.
- **Not a verbatim reproduction of BioDiscoveryAgent.** We use their data and non-essential convention. We *did* run their agent as a controlled comparison (Table 2), but with our backend and model (Haiku 4.5) and small harness adaptations, not their exact published configuration; the controlled 0.105 is close to their reported 0.127 (Sonnet 3.5). Our headline gain over them is a cross-experiment prior they do not have, not a better agent.
- **The B baseline is padded**, on some screens heavily (86% on K562-Essential); it is not a pure-LLM baseline (Table 4†).
- **Genuine ML/LLM agreement is rare** ($n = 101$), so its hit rate carries a wide interval ($46.5\% \pm 9.7$). The *ordering* of the three sources is robust; the exact rate is not.
- **The attribution excludes K562-GWPS**, whose two-stage route lets the LLM make the final selection, so an "ML-only" pick cannot exist there by construction.
- **Calibration is measured only over genes the agent chose to test**, so the score range is narrow by construction and the curve is flat in its interior.
- **Nine screens, five seeds, no prospective wet-lab validation.** The arm is stochastic, which is why every component claim is reported as a *paired* delta.

Appendix: supporting and legacy analyses

These analyses support the main results but are secondary to the paper's central line; the SHAP attributions and the component figures were computed on the legacy predecessor configuration (no detected difference from the final system in aggregate, but not re-estimated under it).

Legacy component decomposition and ablation

These tables quantify each component on the legacy predecessor configuration (not re-estimated under the final system); they motivate the reason-vs-recall analysis in the main text.

Table 10: Stepwise improvement in average hit ratio at round 5 (9 datasets, 5 seeds). Each row introduces one additional component over the previous configuration. The final row is the *legacy* routed C-arm (0.256), used here for the component decomposition; the final leakage-free system (0.253, Table 4[‡]) shows no detected difference in aggregate, but the per-component deltas were not re-estimated under it.

Configuration	Key addition	avg hit@R5	Δ
Random	—	0.066	—
Coreset (A)	Diversity-maximising selection	0.134	+0.068
LOO-LightGBM (static)	Cross-expt. ML prior	0.217	+0.083
OnlineAdaptive	In-expt. ML retraining	0.224	+0.007
LLM branch (B)	LLM naming + static-prior padding	0.225	(*parallel branch*)
C − ML (static LOO + LLM)	Combine LOO prior with LLM	0.247	+0.022 vs. B
Waddington (C, legacy)	Add online ML retraining	**0.256**	+0.009

This legacy stepwise analysis suggests that the cross-experiment prior is the dominant source of performance (+0.083), while LLM endorsement (+0.022 when combined with the LOO prior) and online ML retraining (+0.009) provide smaller increments under that configuration. These quantities are descriptive rather than a formal additive decomposition (the rows add components sequentially, the LLM is a parallel branch, and the comparisons are on the legacy configuration). The explicit cross-experiment memory module provides no measurable gain in this leave-one-out benchmark (paired ablation, Table 11); this is consistent with redundancy with the LLM's parametric knowledge, but could also reflect ineffective retrieval or use of the memory entries.

Table 11: Ablation study, reported as **paired** deltas Δ = variant $-$ C, where C is the full arm *re-run inside the same experiment* (same seeds, same LLM sampling). This matters: the arm is stochastic, so the C baseline re-runs to 0.259–0.262 across the ablation experiments rather than exactly 0.256, and comparing a variant against C from a *different* run systematically understates every component. **C − Mem**: cross-experiment memory removed from the LLM prompt. **C − LLM**: LLM removed; online ML only. **C − ML**: ML frozen at the LOO prior; no in-experiment retraining. **Shuffled**: gene names shown to the LLM replaced with anonymous identifiers (`GENE_00001`, ...). Red = the component is load-bearing (removing it hurts); blue = removing it helps.

Dataset	Δ C − Mem	Δ C − LLM	Δ C − ML	Δ Shuffled
IFN-γ	+0.003	+0.000	−0.013	−0.015
IL-2	−0.013	−0.002	+0.002	−0.005
Sanchez21	+0.004	+0.010	−0.010	−0.008
Sanchez21 (dn)	−0.005	−0.003	−0.010	−0.004
Carnevale22	−0.002	−0.003	+0.006	+0.001
Scharenberg22	+0.012	+0.114	−0.016	+0.078
Steinhart	−0.008	−0.059	−0.004	−0.084
K562-Essential	+0.016	−0.127	−0.025	−0.051
K562-GWPS	−0.004	−0.004	−0.038	−0.004
Average (paired)	**+0.000**	**−0.008**	**−0.012**	**−0.010**

Memory ($\Delta = +0.000$). Removing the cross-experiment memory prompt changes nothing: the paired average is exactly zero. The explicit memory module provided no measurable gain in this setup. This may reflect redundancy with the LLM's parametric knowledge, but ineffective retrieval, presentation, or use of the entries cannot be excluded. DeLM-style verified admission (a mechanical strategy-label check plus an LLM verifier that rejects hallucinated gene names) keeps the retained entries factually grounded.

Online ML retraining ($\Delta = -0.012$). Freezing the ML model at the LOO prior produces the most consistent degradation of any ablation: it hurts on 7 of 9 screens, with the largest losses on K562-GWPS (−0.038) and K562-Essential (−0.025). In the legacy predecessor's paired ablation, removing online retraining produced the largest average loss.

The "LLM" branch ($\Delta = -0.008$ on average, but bimodal) — and an important caveat. C − LLM removes the entire LLM-branch endorsement term from the fusion. That term is *not* purely the LLM: it also carries the static-ML padding described in Table 4†. The two screens with the largest effects are exactly the two with the most padding, so they must be read carefully:

- **K562-Essential (−0.127): this is *not* evidence for LLM biology.** 86% of the branch's genes there come from the static LOO ranker (the LLM names almost nothing that exists in the 623-gene pool). What the ablation removes is the *static cross-experiment prior* being blended into the online model — a real and valuable signal, but not LLM reasoning. We retract the earlier reading of this number.
- **Scharenberg22 (+0.114 when removed):** 68% padding, so "the LLM hurts here" is likewise mostly "the static prior hurts here", conflicting with an unusually strong online ML signal (LOO AUC = 0.825 on K562 DepMap features).

- **Steinhart (-0.059): this one does survive.** Padding is 43%, so a majority of the branch is genuine LLM naming, and the gene-name ablation below independently confirms it.

Steinhart is also where one would *predict* the LLM to matter most, for a reason visible in Table 1: it is the gain-of-function (CRISPRa) screen, asking which genes make CAR-T cells resist exhaustion *when over-expressed*, whereas every DepMap feature we use is derived from *knockout* screens. A gene that is dispensable when deleted can still be potent when over-expressed — IRF4, one of Steinhart's strongest hits, is essential in only 6% of DepMap cell lines — so the ML prior has little to say there, and the LLM's gene-name-conditioned parametric signal appears to fill part of the gap, although we cannot distinguish biological generalization from memorized associations. Consistently, Steinhart has both the lowest hit rate of the nine screens (0.8%) and the largest gene-name-shuffle penalty (-0.084).

Gene-name semantics ($\Delta = -0.010$; -0.084 on Steinhart) — a probe of dependence on gene-name semantics. This ablation is immune to the padding confound in a way C − LLM is not: the branch is still present and still padded, but the LLM can no longer recognise the genes it is naming. Performance collapses on Steinhart (-0.084) and drops on K562-Essential (-0.051), so the LLM's contribution on pathway-specific tasks is mediated by the *gene names* rather than by the task description or by experimental feedback — though whether that name-conditioned signal reflects biological generalization or memorized associations, this ablation cannot say. Scharenberg22 again improves ($+0.078$).

Testing the explanation: is it the gain-of-function modality? (No.)

The story above rests on a sample of one: Steinhart is the only gain-of-function (CRISPRa) screen in the benchmark, so "gain-of-function makes the gene name matter" is a hypothesis fitted to a single point. We tested it. The same study that supplied our IL-2 and IFN-γ screens (Schmidt et al., 2022) ran *both* a CRISPRi and a CRISPRa genome-wide screen for each cytokine, so the CRISPRa arm is the matched gain-of-function counterpart of two screens we already have. We onboarded `IL2_crispra` and `IFNG_crispra` from that arm, keeping everything fixed except the modality: the same anchor-relative features (they are modality-agnostic — activating vs. silencing IL-2 biology shares the same anchors), the same hit convention, an activation-framed task prompt ("genes that, when *over-expressed via CRISPRa*, regulate IL-2 production"), and — because these large screens would otherwise route to the LLM-suppressed `ml_heavy` regime — the same fusion weight as Steinhart ($w_{\text{LLM}} = 0.4$).

If gain-of-function were the cause, the gene name should be worth $\approx +0.08$ here too. It is not:

Screen (gain-of-function, CRISPRa)	name value (A2−A1)	cf. its CRISPRi twin
`IL2_crispra`	$+0.002 \pm 0.003$	-0.050
`IFNG_crispra`	-0.004 ± 0.001	$+0.013$
Steinhart (reference)	$+0.080 \pm 0.013$	—

On both new gain-of-function screens the gene name is worth nothing, an order of magnitude below Steinhart. **Perturbation modality does not explain Steinhart's name-dependence.** The corrected-prompt run barely moved these numbers from a first run with a placeholder prompt, so this is not an artefact of under-specifying the task; the name simply does not help here.

What *does* distinguish Steinhart is the shape of its hit set: 145 concentrated hits (0.8%) dominated by canonical exhaustion regulators (IRF4, TOX, . . .) with highly recognizable gene-name semantics, against ∼900 diffuse top-movers (∼5%) on each CRISPRa cytokine screen, where recognizable names are a small minority. Our loss-of-function-trained prior also transfers poorly to CRISPRa hits (hit@R5 ≈ 0.12 vs. 0.35 on the CRISPRi twin), so the whole system is weak there. We leave the hit-set-concentration hypothesis untested. The operational consequence is clear enough, though: **we do not route by modality** — the structure-only reasoning mode stays an opt-in choice, not a default triggered by whether a screen is CRISPRa.

An accidental confirmation: the feature router already knew

The C-arm selects its feature set per screen. That router was tuned *empirically*, by trying feature sets and keeping whatever scored best, long before we understood why. It converged on excluding **all four DepMap features** for exactly two screens — Steinhart and K562-Essential — because including them *hurt*. The code comment recording this guessed at the reason ("DepMap disrupts online learning for GD2 biology"), and that guess was wrong: Steinhart does not measure GD2 biology at all (Table 1).

The real reason is the one above. **Every DepMap feature is derived from knockout screens**, and Steinhart is the one *gain-of-function* (CRISPRa) screen: it asks which genes make CAR-T cells resist exhaustion *when over-expressed*. A gene that is dispensable when deleted can be potent when over-expressed — IRF4, among Steinhart's strongest hits, is essential in only 6% of DepMap cell lines — so a knockout-derived prior is not merely uninformative there, it is actively misleading, and the router learned to discard it.

Reading the SHAP attributions back out confirms that this is what the model actually does:

Table 12: The ML model's most-used features, per screen (mean |SHAP| over the genes it selected). DepMap essentiality dominates wherever it is available — and is absent exactly where a knockout-derived prior does not transfer.

Screen	Perturbation	Top features used by the ML model
IFN-γ, IL-2, Carnevale22, Scharenberg22, K562-GWPS	loss-of-function	**depmap_frac_ess** (rank 1), then PPI / pLI
Steinhart	**gain-of-function** (CRISPRa)	PPI, STRING degree, pathway counts — **no DepMap feature at all**
K562-Essential	loss-of-function	PPI, STRING degree — **no DepMap feature**

This offers a plausible explanation for why Steinhart is the screen where the LLM matters most: it is precisely the screen where the ML model's single strongest feature family has been removed, leaving it with weak topological signal — and a gene-name-conditioned parametric signal appears to fill part of the gap, although its origin cannot be distinguished from memorized associations. (K562-Essential drops DepMap for a different reason: it *is* a curated essentiality screen, so a pan-cancer essentiality prior is close to a restatement of the label and adds nothing.)

A freely planning tool-using agent underperforms the constrained pipeline

The natural next step from a pipeline is a *tool-using agent* in the BioDiscoveryAgent / PerTurboAgent mould: let the LLM plan, call an ML-ranking tool and a pathway-enrichment tool, and commit a batch itself. We implemented exactly that (a task-specialised harness with a JSON action protocol; `ml_rank` / `enrich` / `finish`; no agent framework) and benchmarked it against the pipeline on identical screens and budgets. We state the scope up front: this is one action set, one prompt, one stopping rule and one budget, so the result below is evidence that *a freely planning agent does not beat the constrained fusion policy here*, not that tools are useless in general — a different action schema or a policy that cannot override a confident ML score could close the gap.

Table 13: Tool-using agent vs. the fixed-policy constrained pipeline (hit ratio @ R5). The agent plans its own rounds and calls tools; the pipeline fuses online ML with constrained LLM endorsement under a fixed policy. The agent is *worse on 8 of 9 screens*.

Dataset	Tool-using agent	Pipeline (C)	Δ
Scharenberg22	**0.582**	0.465	+0.116
Carnevale22	0.044	**0.056**	−0.012
IFN-γ	0.163	**0.190**	−0.027
Sanchez21	0.059	**0.087**	−0.029
Sanchez21 (dn)	0.049	**0.099**	−0.050
Steinhart	0.100	**0.159**	−0.059
IL-2	0.264	**0.347**	−0.083
K562-GWPS	0.209	**0.347**	−0.138
K562-Essential	0.413	**0.556**	−0.143
Average	0.209	**0.256**	**−0.047**

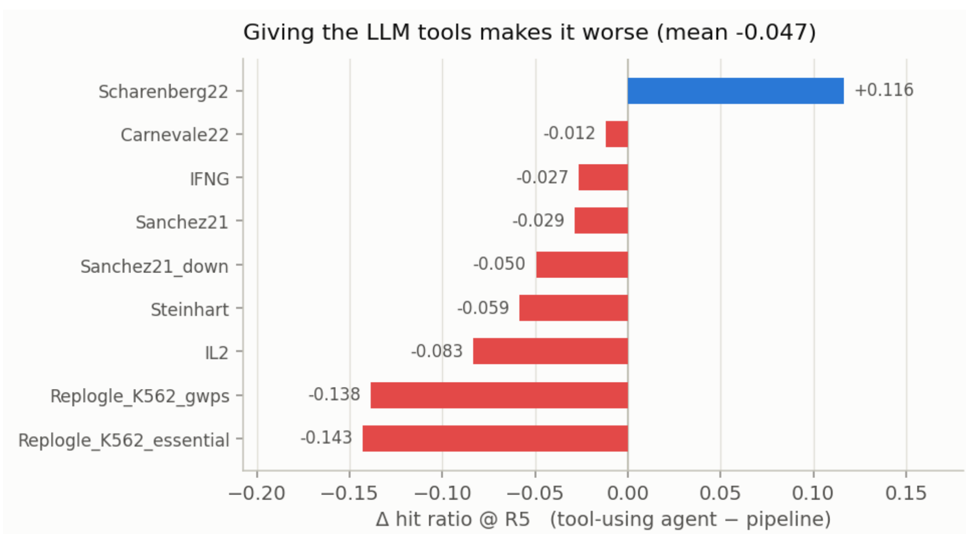

Figure 6: Per-screen delta of the tool-using agent against the pipeline. It wins on exactly one screen and loses worst precisely where the ML signal is strongest (K562-Essential, K562-GWPS) — given freedom, the LLM *overrides* a well-calibrated model.

The agent wins on exactly one screen (Scharenberg22, +0.116), where its runtime pathway-enrichment tool pays off, and loses most on the two genome-wide screens where the ML prior is strongest. The failure mode is instructive: given the freedom to act, the LLM overrides a model that is better calibrated than it is. **Constraining the LLM to a fixed fusion policy is not a limitation of the pipeline — it appears to be the reason the pipeline wins under this tested harness** (one action schema, prompt, stopping rule, and budget).

Transplanting the agent's one good idea

Since enrichment was the agent's only advantage, we transplanted it into the pipeline (Enrichr over the hits revealed so far, injected into the LLM prompt each round, gated to significant and coherent pathways), and separately tested an externalised *skill library* distilled from past experiments.

Table 14: Agent-style augmentations of the pipeline, as **paired** deltas over the same runs. Both are within noise of zero.

Augmentation	C	C + augmentation	Δ (paired)
Runtime pathway enrichment (gated)	0.260	0.263	+0.003
Externalised skill library	0.258	0.259	+0.001

Runtime enrichment captures the Scharenberg22 gain (+0.061 on that screen) without the collateral damage the free agent suffered, but nets only +0.003 overall: on genome-wide essentiality screens the enriched terms are highly significant yet *non-selective* (ribosome, proteasome, spliceosome), so "prioritise genes in these pathways" barely narrows a 9,000-gene pool. The skill library adds +0.001. Together with the memory ablation (+0.000), this is a consistent negative result about *externalisation*: memory, skills and retrieved pathways all add ≈ 0 once the model already has its parametric knowledge plus the hits it has observed.